



PSO201/PHY204: Assignment 5

Jan-Apr 2025

1. Consider the spherical harmonics given below.

$$Y_{00} = \frac{1}{\sqrt{4\pi}}; Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta; Y_{11} = -\sqrt{\frac{3}{4\pi}} \sin \theta e^{i\varphi}; Y_{21} = -\sqrt{\frac{15}{8\pi}} e^{i\varphi} \sin \theta \cos \theta$$

Show that these functions are eigenfunctions of both L^2 and L_z , and compute the corresponding eigenvalues.

2. Consider a particle in a normalized eigenstate of L^2 and L_z , where $\Psi \propto Y_{lm}$

a) Find $\langle L_x \rangle, \langle L_y \rangle$. Find $\langle L_x^2 \rangle$ and $\langle L_y^2 \rangle$.

b) Using the results, find $\Delta L_x \Delta L_y$.

3. A quantum-mechanical state of a particle is described by the normalized wave function $\psi(x, y, z) = \frac{\alpha^{5/2}}{\sqrt{\pi}} z e^{-\alpha r}$. Show that the system is in a state of definite angular momentum and find the values of L^2 and L_z , associated with the state.

4. Consider four Hermitian 2×2 matrices, $I, \sigma_1, \sigma_2, \sigma_3$ where I is the unit matrix, and the others satisfy the relation $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} I$. Prove the following without using a concrete representation for the matrices.

a) Prove that $\text{Tr}(\sigma_i) = 0$.

b) Show that the eigenvalues of σ_i are ± 1 and that $\det(\sigma_i) = -1$.

c) Given that $A = a_0 I + \sum_{i=1}^3 a_i \sigma_i$; where A is any 2×2 matrix, show that $a_i = \frac{1}{2} \text{Tr}(\sigma_i A)$

5. The eigenvectors for a free electron spin in a basis where $\hat{S}_z$ is diagonal can be written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with eigenvalues of $\hat{S}_z$ being $+1/2$ and $-1/2$ (in units of $\hbar$) respectively. Using this basis, find a normalized eigenvector of $\hat{S}_y$, with eigenvalue $-1/2$ (in units of $\hbar$).

6. Construct a matrix representing $\hat{S}_{\hat{n}}$, the component of spin angular momentum along an arbitrary direction $\hat{n}$. Find the eigenvalues and eigenvectors of $\hat{S}_{\hat{n}}$.

7. A spin $1/2$ particle of mass m and charge $-e$ is subjected to a magnetic field $\vec{B}$ along z -direction.

a) If the initial state is $|\uparrow\rangle$ at $t = 0$, find the state at a later time t .

b) If the particle is in the state $|S_x; +\rangle$ at $t = 0$, find the probability for the particle to be in the same state at a later time t .

c) Now if $\vec{B} = B\hat{x}$, find the probability of the particle, in the initial state $|\uparrow\rangle$, to remain in the same state at a later time t .

8. Find out the matrix representation of Spin along an arbitrary direction $\hat{n}$ and the corresponding eigenstates in terms of the orthonormal basis vectors $|\uparrow\rangle, |\downarrow\rangle$. Consider a spin half particle in the initial state $|\uparrow\rangle$ at $t = 0$. A magnetic field B is applied in the arbitrary direction $\hat{n}$. Calculate the probability of finding the particle to be in $|\downarrow\rangle$ state at a later time t .

9. An electron is in the spin state $A \begin{pmatrix} 3i \\ 4 \end{pmatrix}$. Determine the normalization constant. Find $\langle S_x \rangle, \langle S_y \rangle, \langle S_z \rangle; \langle S_x^2 \rangle, \langle S_y^2 \rangle, \langle S_z^2 \rangle$; and the corresponding uncertainties.

10. Consider a system of spin $1/2$. Find the eigenvalues and normalized eigenvectors of the operator $A\hat{S}_y + B\hat{S}_z$, where $\hat{S}_y, \hat{S}_z$ are the angular momentum operators, and A and B are real constants.